

INDIAN INSTITUTE OF TECHNOLOGY ROORKEE

ROORKEE – 247667

Database Management Systems (CSC-301)

Sheet: 1

Total Marks: 100

Instructions:

- The deadline for assignment submission is **17 August till 11:59 PM**.
- If there is any similarity between the two students' submissions, both will be awarded **zero marks**. So, it's the student's responsibility not to share the submission with other students.
- Each late submission will receive a 15% penalty per day for up to 2 days. No submission will be accepted after the 2nd late day.
- You must handwrite your solutions; please scan the hard copy and convert it into a PDF with your name and enrolment number. Your name and roll number should be clearly written on the top of the first page.
- Please do not email us your submissions.
- Your submission must be uploaded to the Microsoft team.
- For questions five to seven, mention all the steps clearly; otherwise, you will lose marks.

Que. 1: What are the advantages of a DBMS over a file processing system? When would it make sense not to use a database system? [15 Marks]

Que. 2: Explain the difference between the view(external), physical(internal), and logical (conceptual) schemas. How are these different schema layers related to the concept of logical and physical data independence? [15 Marks]

Que. 3: List the reasons why null values might be introduced into the database? [5 Marks]

Que. 4: What are the design goals for relational databases, and explain why each is desirable? [15 Marks]

Que. 5: Let R be a relation schema and F be a set of functional dependencies on R . Find all Candidate Key(s) for relation R ? [25 Marks]

- i. $R(ABCDE)$
 $F = \{AB \rightarrow C, C \rightarrow D, D \rightarrow E, A \rightarrow B, C \rightarrow A\}$
- ii. $R(ABCDE)$
 $F = \{A \rightarrow D, AB \rightarrow C, B \rightarrow E, D \rightarrow C, E \rightarrow A\}$
- iii. $R(ABCDE)$
 $F = \{AB \rightarrow C, C \rightarrow D, D \rightarrow E, E \rightarrow A, D \rightarrow B\}$
- iv. $R(ABCDEF)$
 $F = \{AB \rightarrow C, C \rightarrow DE, E \rightarrow F, F \rightarrow A\}$
- v. $R(ABCDEFGH)$
 $F = \{AB \rightarrow CD, D \rightarrow EG, F \rightarrow H, C \rightarrow EF, H \rightarrow A, G \rightarrow B, A \rightarrow B\}$

Que. 6: Let $R(ABCDEFGH IJ)$ be a relation schema and $F = \{AB \rightarrow C, A \rightarrow DE, B \rightarrow F, F \rightarrow GH, D \rightarrow IJ\}$ be a set of functional dependencies on R . Determine the schema is lossless or lossy for the following decomposition? [15 Marks]

- i. $D1 = \{ABC, ADE, BF, FGH, DIJ\}$
- ii. $D2 = \{ABCDE, BFGH, DIJ\}$
- iii. $D3 = \{ABCD, DE, BF, FGH, DIJ\}$

Que. 7: Let $R(ABCDEG)$ be a relation schema and $F = \{AB \rightarrow C, AC \rightarrow B, AD \rightarrow E, B \rightarrow D, BC \rightarrow A, E \rightarrow G\}$ be a set of functional dependencies on R . Determine the following decomposition is dependency preserving or not? [10 Marks]

- i. $D1 = \{ABC, ACDE, ADG\}$
- ii. $D2 = \{ABC, ADEG\}$